

ALLEN AAF, AK, USA

WMO: 702670

Lat: 63.994N

Lon: 145.721W

Elev: 1277

StdP: 14.03

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26415

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.9	-30.4	-33.3	0.9	-27.1	-30.0	1.1	-23.7	41.7	2.8	39.1	4.6	2.1	170	0.765

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	78.5	58.8	75.1	57.0	72.0	56.0	60.9	74.2	59.3	71.2	57.6	68.4	8.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.8	69.5	63.1	54.1	65.3	61.7	52.5	61.5	60.7	27.5	74.8	26.3	71.2	25.3	68.4	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.1	30.5	26.8	DB	-42.6	83.4	5.4	3.3	-46.5	85.8	-49.6	87.7	-52.7	89.6	-56.6	92.0	(4)
(5)			WB	-32.6	63.8	3.2	4.3	-34.9	67.0	-36.8	69.5	-38.5	71.9	-40.8	75.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.5	0.0	8.5	14.7	33.8	48.4	58.2	60.3	55.5	45.1	27.6	8.5	3.5	
	DBStd	24.80	18.03	15.67	14.11	10.29	7.90	5.86	5.20	6.42	8.11	11.11	14.13	15.26	
	HDD50	8010	1549	1161	1093	488	125	5	3	19	183	695	1246	1443	
	HDD65	12640	2014	1581	1558	936	515	215	158	303	598	1158	1696	1908	
	CDD50	876	0	0	0	1	75	252	323	188	35	2	0	0	
	CDD65	35	0	0	0	0	1	12	14	8	0	0	0	0	
	CDH74	388	0	0	0	0	19	131	146	92	0	0	0	0	
	CDH80	49	0	0	0	0	1	20	16	12	0	0	0	0	
Wind	WSAvg	9.6	12.3	11.9	9.9	9.1	8.9	7.5	7.2	7.0	8.5	9.5	10.9	12.6	
Precipitation	PrecAvg	11.5	0.3	0.2	0.2	0.2	0.9	2.3	2.8	2.0	1.0	0.6	0.5	0.3	
	PrecMax	17.7	1.1	0.7	1.1	0.7	3.1	5.4	6.0	4.8	1.9	2.1	1.4	1.0	
	PrecMin	7.7	0.0	0.0	0.0	0.0	0.1	0.3	0.7	0.6	0.2	0.0	0.0	0.0	
	PrecStd	2.9	0.3	0.2	0.2	0.2	0.7	1.3	1.3	0.9	0.5	0.5	0.4	0.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.2	43.5	48.7	63.7	76.4	83.1	82.4	81.9	67.2	58.0	41.6	42.9	
		MCWB	37.1	35.6	38.8	46.6	53.2	59.4	61.0	60.4	52.1	47.2	35.4	36.0	
	2%	DB	36.0	37.3	42.9	56.8	71.4	77.7	78.1	76.5	63.7	50.5	33.7	33.6	
		MCWB	30.7	30.8	34.7	42.7	51.2	57.4	59.1	59.1	50.0	41.6	30.6	29.2	
	5%	DB	28.7	32.3	39.1	52.2	67.1	74.1	74.8	71.5	60.8	46.6	30.0	27.3	
		MCWB	25.7	28.1	32.0	39.9	49.5	56.1	58.1	56.9	48.8	38.9	27.4	24.5	
	10%	DB	23.7	27.7	34.9	48.9	62.9	70.5	71.8	67.4	58.1	42.1	26.3	23.4	
		MCWB	21.5	24.4	28.9	38.0	47.6	54.5	57.1	54.9	47.5	36.5	24.1	21.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.6	36.1	39.1	47.4	55.0	62.4	63.1	62.7	53.1	47.5	35.3	36.4
			MCDB	43.7	43.6	48.6	61.9	71.3	79.0	77.4	78.7	63.7	57.6	41.7	42.3
2%		WB	31.1	31.5	35.0	43.4	52.7	59.5	61.0	60.4	51.2	42.2	31.2	29.7	
		MCDB	36.3	37.0	42.1	56.2	68.2	73.2	73.0	73.6	60.7	49.4	33.7	33.4	
5%		WB	26.3	28.4	32.3	40.5	50.6	57.6	59.6	58.1	50.0	39.3	27.6	24.7	
		MCDB	29.2	32.7	38.8	51.1	65.2	70.1	71.0	68.8	58.7	45.7	30.0	27.4	
10%		WB	22.0	24.6	29.2	38.4	48.4	55.7	58.3	56.0	48.5	36.8	24.1	21.3	
		MCDB	24.2	27.9	34.9	48.0	61.6	67.8	69.1	64.8	56.7	41.7	26.3	23.5	
Mean Daily Temperature Range		5% DB	MDBR	12.4	14.4	19.2	18.6	18.3	17.3	16.0	16.3	15.5	12.0	11.8	12.1
			MCDBR	15.8	15.2	18.3	22.9	24.6	23.3	22.8	23.1	19.8	14.8	13.3	16.7
	MCWBR		12.5	11.1	12.5	12.8	11.2	9.6	9.5	10.3	10.3	8.9	10.4	13.3	
	5% WB	MCDBR	15.8	14.9	17.7	21.8	22.3	20.2	19.0	20.5	17.6	13.7	13.0	17.1	
		MCWBR	12.6	11.1	12.1	12.6	10.2	8.9	8.6	9.8	9.7	8.6	10.1	13.9	
Clear-Sky Solar Irradiance	taub	0.203	0.266	0.288	0.336	0.359	0.378	0.381	0.390	0.306	0.274	0.199	0.198		
	taud	1.938	2.122	2.180	2.240	2.334	2.316	2.341	2.297	2.521	2.387	1.970	1.910		
	Ebn at Noon	116	204	253	265	270	266	261	245	253	209	115	45		
	Edh at Noon	9	20	28	34	34	36	34	32	21	15	8	4		
All-Sky Solar Radiation	RadAvg	59	279	838	1457	1694	1713	1495	1180	735	323	90	23		
	RadStd	6	25	56	73	109	134	101	98	76	36	10	1		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	N/A
(47) Regional (5 neighbors)	+1.55	N/A	N/A	N/A	N/A	N/A	N/A	-587	-637	N/A	N/A

Nomenclature: See separate page